

Spectral Norms of Independent-Entry Matrices with Regular Moment Growth

Diar Heidary

Research draft — 14 September 2026

Attribution revision: 16 September 2026

AI-assisted research draft. AI tools assisted the mathematical derivation, checking, and exposition under the author’s direction. The argument has undergone internal analytic reviews and bounded exact checks during drafting. These are not external peer review or, by themselves, proof-assistant verification. A separate public Lean record for the upper comparison is identified in Section 10; it must not be conflated with verification of every analytic argument in this draft. Established results used below are stated and credited.

Abstract

We establish the expected spectral-norm estimate in MI-32 for real matrices with independent mean-zero entries whose moments grow regularly. The comparison uses the largest row and column variance sums and the original shared-index-deletion parameter built from weak bilinear moments at order $\log(k + 1)$, including the exponent below one at $k = 1$. The local upper estimate follows by representing the original entries as independent signs times their magnitudes. Sign parity gives orthogonal sectors, while every magnitude power stays in the polynomial coefficients. Creation localizes a small set of original rows; annihilation localizes a small set of original columns. A positive original-coefficient polynomial of degree d satisfies an L_2 -to- L_4 bound with factor $(\sqrt{3}\alpha)^d$, proved by Cauchy–Schwarz, moment doubling, and a direct Hilbert-valued Boolean argument. A net on the small axis and the standard strong/weak moment theorem then bound each sector before its squared norm is summed. The actual vacuum walk polynomials remain in this cone at every step. We give the subsequent deletion and symmetrization reductions with their precise index and moment costs, and present the established lower comparison with its source attribution. No unrestricted polynomial-space norm or matching quantile statement is asserted.

Contents

1	Introduction and the exact mean-norm statement	3
1.1	The proof mechanism	4
2	Linear moments and a Boolean polynomial estimate	4
2.1	Strong and weak moments of a linear image	4
2.2	The Hilbert Boolean estimate, with a direct proof	5
3	Positive original coefficients control polynomial moments	5
4	The exact parity–amplitude representation	6
4.1	The space and its polynomial domain	7
4.2	Multiplication, creation, and annihilation	7
5	Exact localization by the two original count vectors	8
6	Thin matrices and the cone contraction	9
7	Vacuum walks and the local moment theorem	10
8	From the local estimate to shared-index deletion	11
8.1	Scalar comparison, including the subunit endpoint	11
8.2	The local estimate in the matrix-symmetric convention	11
8.3	The established variance-envelope input and its specialization	12
8.4	Nested deterministic deletions and screened coefficients	13
8.5	Two block families and their maximum norms	14
8.6	The remaining entries have a summable variance envelope	15
8.7	The actual dilation and its paired deletion budget	15
8.8	Symmetrization without changing the small exponent	16
9	The known lower bound on the same matrix	16
9.1	Row and column variances	16
9.2	A large weak moment has a small original support	16
9.3	A uniform scalar lower tail	17
9.4	Disjoint witnesses for each deletion budget	17
10	Scope and the information retained by the proof	18

1 Introduction and the exact mean-norm statement

Let $X = (X_{ij})_{i,j \leq n}$ be a real matrix with independent mean-zero entries. We assume that every absolute moment is finite and that one constant $\alpha \geq 1$ satisfies

$$\|X_{ij}\|_{L_{2p}} \leq \alpha \|X_{ij}\|_{L_p}, \quad p \geq 1, \quad i, j \in [n]. \quad (1)$$

Here $[n] = \{1, \dots, n\}$ and $\|Z\|_{L_p} = (\mathbb{E}|Z|^p)^{1/p}$ for every $p > 0$. At $p < 1$ this notation denotes a moment functional, not a norm. All logarithms below are natural. Matrix norms without a subscript are operator norms for the Euclidean norms on domain and range. For a scalar random variable we also use the abbreviation $\|Z\|_p = \|Z\|_{L_p}$; for a deterministic vector, the subscript 2 denotes its Euclidean norm.

Define the deterministic variance scales

$$\begin{aligned} \sigma_r(X) &= \max_i \left(\sum_j \mathbb{E} X_{ij}^2 \right)^{1/2}, \quad \sigma_c(X) = \max_j \left(\sum_i \mathbb{E} X_{ij}^2 \right)^{1/2}, \\ M(X) &= \sigma_r(X) + \sigma_c(X), \quad \sigma(X) = \max\{\sigma_r(X), \sigma_c(X)\}. \end{aligned} \quad (2)$$

For a deterministic set $I \subset [n]$ and $p > 0$, put

$$R_{X,I}(p) = \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum_{i,j \notin I} X_{ij} s_i t_j \right\|_{L_p}, \quad R_X(p) = R_{X,\emptyset}(p). \quad (3)$$

The vectors in this supremum are deterministic. The deletion parameter is

$$D(X) = \max_{1 \leq k \leq n} \min_{I \subset [n], |I| \leq k} R_{X,I}(\log(k+1)). \quad (4)$$

One and the same original set I deletes both rows and columns. It is chosen for the law, before any realization is observed. The minimum is outside the vector supremum and is attained, since the collection of sets is finite. Deleting all indices gives zero, so the $k = n$ term is zero. The $k = 1$ term retains its exact exponent $\log 2 < 1$.

Theorem 1.1 (Independent entries with regular moment growth). *For every $\alpha \geq 1$ there are constants $c_\alpha, C_\alpha > 0$ such that every $n \geq 1$ and every matrix satisfying (1) obey*

$$c_\alpha \{M(X) + D(X)\} \leq \mathbb{E}\|X\| \leq C_\alpha \{M(X) + D(X)\}. \quad (5)$$

The constants depend only on α , not on the dimension, the individual entry distributions, or the deterministic deletion sets.

The target is Conjecture 4.3 of Latała–Świątkowski, recorded as problem MI-32 [4, Conjecture 4.3][1]. The lower comparison is their Theorem 4.1, with the sparsification step supplied by their Lemma 4.2. Their Remark 4.5 identifies the local-to-global deletion route used here. These are distinct prior contributions; the proof below reproduces the needed reductions rather than claiming them as new.

The contribution developed here is the local upper estimate obtained by retaining sign parity, ordinary polynomial degree, and the nonnegative coefficients of actual matrix walks. The deletion argument preserves the original shared-index convention and states its external probabilistic inputs explicitly. In particular, this version quotes the independent-coordinate strong/weak theorem of Latała–Strzelecka [2, Theorem 1.1] and a variance-envelope estimate of Latała–van Handel–Youssef [5, Theorem 4.4].

Related cases and later comparison bounds. Latała–Świątkowski also prove the Gaussian-mixture case in Proposition 4.4 [4]. For weighted signs, Latała’s Theorem 1.1 treats zero–one coefficient matrices, whereas Theorem 1.9 treats arbitrary deterministic coefficients with a triple-logarithmic upper loss [11]. These two statements should not be interchanged. Meller’s Theorem 1.1 supplies a polyloglog-loss comparison for regular symmetric entries [12]. Its displayed weak-moment parameter is undeleted, and logarithm and normalization conventions in these comparisons are not literally (4). They are cited as prior comparison results, not silently substituted for the parameter used in this paper. No priority or exhaustive novelty claim is made here.

1.1 The proof mechanism

For symmetric entries, write $Y_e = \varepsilon_e Z_e$ in law, where $Z_e = |Y_e|$ and the original signs are independent of all magnitudes. The signs give an orthogonal parity basis. Magnitude powers stay inside its coefficients, so visiting an entry repeatedly does not lose its moment information. In the row-to-column direction, parity creation has exact sectors involving few original rows, whereas parity annihilation has different exact sectors involving few original columns. A net on this small axis has constant cost at the relevant moment order.

The remaining estimate is a polynomial moment comparison on the cone of nonnegative original monomial coefficients. That cone contains every vacuum walk polynomial and is preserved by the parity projections and every subsequent original multiplication. It is not the entire linear span of these polynomials. This distinction permits the argument to retain rare-amplitude effects without bounding an unrestricted orthogonal-polynomial multiplication operator.

The standard inputs are Boolean hypercontractivity and the independent-coordinate strong/weak moment comparison. The final deletion argument also states the variance-envelope estimate it uses. These public theorems are cited in their applicable forms. All representations, cone estimates, count decompositions, and MI-32 reductions are given in this manuscript.

2 Linear moments and a Boolean polynomial estimate

2.1 Strong and weak moments of a linear image

The following is the finite-dimensional normed-space formulation of [2, Theorem 1.1]. It follows from that theorem by taking the index set to be the dual unit ball.

Theorem 2.1 (Independent-coordinate strong/weak comparison). *Let $W_1, \dots, W_N$ be independent centered variables with finite moments and $\|W_e\|_{L_{2p}} \leq \alpha \|W_e\|_{L_p}$ for $p \geq 2$. For deterministic vectors x_e in a finite-dimensional normed space V , and every $p \geq 1$,*

$$\begin{aligned} \left(\mathbb{E} \left\| \sum_e W_e x_e \right\|_V^p \right)^{1/p} &\leq C_{\text{SW}}(\alpha) \left[\mathbb{E} \left\| \sum_e W_e x_e \right\|_V \right. \\ &\quad \left. + \sup_{\|\phi\|_{V^*} \leq 1} \left\| \sum_e W_e \phi(x_e) \right\|_{L_p} \right]. \end{aligned} \tag{6}$$

The constant depends only on α .

In particular this applies to a Hilbert norm of a fixed linear image; there is no requirement that its vector coordinates themselves be independent. We will use it first with the independent entries as the W_e and a deterministic row vector as part of the coefficients x_e . It does not supply an estimate for the matrix mean norm unless that mean has been independently bounded.

2.2 The Hilbert Boolean estimate, with a direct proof

On independent signs $\varepsilon_1, \dots, \varepsilon_N$, write $\varepsilon_S = \prod_{e \in S} \varepsilon_e$ and let N_ρ be the noise operator defined on Fourier characters by $N_\rho \varepsilon_S = \rho^{|S|} \varepsilon_S$. The L_2 -to- L_4 case of classical Boolean hypercontractivity is sufficient for this version of the argument. We give its Hilbert-valued proof for completeness. The historical sources are Bonami and Beckner [6, 7]; the general Boolean formulation used for reference is stated in O'Donnell's Lecture 16, Theorem 1.1 [3]. Re-proving this special case does not make the hypercontractive inequality itself a new result.

Lemma 2.2 (Hilbert-valued Boolean polynomials). *If $F = \sum_{|S| \leq d} \varepsilon_S v_S$ takes values in a real Euclidean space, then*

$$\|F\|_{L_4(\ell_2)} \leq 3^{d/2} \|F\|_{L_2(\ell_2)}. \quad (7)$$

Proof. For fixed real Hilbert vectors a, b ,

$$\mathbb{E}_\varepsilon \|a + \varepsilon b\|_2^4 = (\|a\|_2^2 + \|b\|_2^2)^2 + 4\langle a, b \rangle^2 \leq \|a\|_2^4 + 6\|a\|_2^2 \|b\|_2^2 + \|b\|_2^4.$$

If A, B depend on other signs, integrate this inequality and use Cauchy–Schwarz in the mixed term. Taking a square root gives

$$\|A + \varepsilon B\|_{L_4(\ell_2)}^2 \leq \|A\|_{L_4(\ell_2)}^2 + 3\|B\|_{L_4(\ell_2)}^2. \quad (8)$$

Set $\rho = 1/\sqrt{3}$ and induct on the number of signs. Write $F = A + \varepsilon_N B$, where A, B depend on the first $N - 1$ signs. The noise operator on those signs is denoted by N'_ρ . The induction hypothesis and (8) yield

$$\|N_\rho F\|_{L_4(\ell_2)}^2 \leq \|N'_\rho A\|_{L_4(\ell_2)}^2 + 3\rho^2 \|N'_\rho B\|_{L_4(\ell_2)}^2 \leq \|A\|_{L_2(\ell_2)}^2 + \|B\|_{L_2(\ell_2)}^2 = \|F\|_{L_2(\ell_2)}^2.$$

The base case of zero signs is equality. Finally set $G = \sum_{|S| \leq d} \rho^{-|S|} \varepsilon_S v_S$, so $N_\rho G = F$. Hilbert Fourier orthogonality gives

$$\|F\|_{L_4(\ell_2)} \leq \|G\|_{L_2(\ell_2)} = \left(\sum_{|S| \leq d} \rho^{-2|S|} \|v_S\|_2^2 \right)^{1/2} \leq \rho^{-d} \|F\|_{L_2(\ell_2)}.$$

□

3 Positive original coefficients control polynomial moments

Fix independent symmetric variables $W = (W_e)_{e \in N}$ satisfying (1), and put $Z_e = |W_e|$. For an exponent vector $\nu \in \mathbb{N}_0^N$, write $Z^\nu = \prod_e Z_e^{\nu_e}$ and $|\nu| = \sum_e \nu_e$. Polynomial degree below always means this ordinary total degree, including all repeated powers.

Lemma 3.1 (A product-moment bound). *For $m_e(u) = \mathbb{E} Z_e^u$, $u \in \mathbb{N}_0$, and $m_e(0) = 1$,*

$$m_e(u + v) \leq \alpha^{u+v} m_e(u) m_e(v), \quad u, v \in \mathbb{N}_0. \quad (9)$$

Consequently, every scalar polynomial $G(Z) = \sum_\nu a_\nu Z^\nu$ with $a_\nu \geq 0$ and degree at most D satisfies

$$\mathbb{E} G(Z)^2 \leq \alpha^{2D} (\mathbb{E} G(Z))^2. \quad (10)$$

Proof. For positive integers u, v , Cauchy–Schwarz and moment doubling give

$$\begin{aligned} m_e(u + v) &\leq \sqrt{m_e(2u) m_e(2v)}, \\ m_e(2u) &= \|W_e\|_{L_{2u}}^{2u} \leq \alpha^{2u} \|W_e\|_{L_u}^{2u} = \alpha^{2u} m_e(u)^2. \end{aligned}$$

Apply the second formula also with v to prove (9). If either exponent is zero, the assertion follows from $\alpha \geq 1$; this does not apply moment doubling at zero. Identically zero variables cause no exception.

For two monomials of G , independence gives

$$\mathbb{E}Z^{\nu+\mu} = \prod_e m_e(\nu_e + \mu_e) \leq \alpha^{|\nu|+|\mu|} \prod_e m_e(\nu_e) \prod_e m_e(\mu_e).$$

Multiply by $a_\nu a_\mu \geq 0$, sum over ν, μ , and use $|\nu| + |\mu| \leq 2D$ to obtain (10). $\square$

Lemma 3.2 (Positive-coefficient polynomial comparison). *Let $F(W)$ be a polynomial taking values in a fixed finite-dimensional real Euclidean space. Suppose every coordinate has nonnegative coefficients in the ordinary monomial basis of the original W_e , and $\deg F \leq d$. With*

$$H_\alpha = \sqrt{3}\alpha, \tag{11}$$

one has

$$\|F\|_{L_4(\ell_2)} \leq H_\alpha^d \|F\|_{L_2(\ell_2)}, \tag{12}$$

$$\|F\|_{L_{2p/(p-2)}(\ell_2)} \leq H_\alpha^{4d/p} \|F\|_{L_2(\ell_2)}, \quad p \geq 4. \tag{13}$$

Proof. Represent $W_e = \varepsilon_e Z_e$ in law with independent signs and independent magnitudes. Group ordinary monomials by their exponent parity:

$$F(\varepsilon Z) = \sum_S \varepsilon_S F_S(Z).$$

Every coordinate of each F_S still has nonnegative coefficients. Its monomials have exponent parity S , degree at most d , and hence $|S| \leq d$. Conditional on Z , Lemma 2.2 gives

$$\mathbb{E}_\varepsilon \|F(\varepsilon Z)\|_2^4 \leq 3^{2d} \left(\sum_S \|F_S(Z)\|_2^2 \right)^2. \tag{14}$$

The scalar polynomial $G(Z) = \sum_S \|F_S(Z)\|_2^2$ has nonnegative coefficients and degree at most $2d$. Conditional sign orthogonality gives $\mathbb{E}G = \mathbb{E}\|F\|_2^2$. Thus (10), with $D = 2d$, proves

$$\mathbb{E}\|F\|_2^4 \leq 3^{2d} \alpha^{4d} (\mathbb{E}\|F\|_2^2)^2.$$

Its fourth root is (12). For $u = 2p/(p-2) \in [2, 4]$, the interpolation identity $1/u = (1 - 4/p)/2 + (4/p)/4$ proves (13). Degree zero gives equality in both comparisons. $\square$

The hypothesis concerns coefficients, not pointwise positivity of $F(W)$. A coordinate such as $W_1 + W_2$ is allowed although its value may be negative. Nor does the lemma apply to every polynomial merely because the law is symmetric: arbitrary negative monomial coefficients may create cancellations in the L_2 norm. We will only project or multiply polynomials in ways that preserve the stated coefficient cone.

4 The exact parity–amplitude representation

The sign basis below is the classical Walsh–Fourier basis on a product space [8]. More generally, representations of moment sequences by vacuum matrix elements have an established interacting-Fock-space history [9]. Only elementary multiplication on the original probability space is needed here. The representation is distinguished from the estimates on its preserved positive-coefficient domain.

For now let $Y = (Y_{ij})_{i,j \leq n}$ have independent symmetric entries satisfying (1). There is one original label $e = (i, j)$ for each nonzero entry; denote this finite label set by $\mathcal{E}$. Keeping zero entries as zero multipliers would give the same formulas. Row vertices r_i and column vertices c_j belong to disjoint copies of $[n]$; write $\mathcal{V}$ for their union.

Represent the joint law as $Y_e = \varepsilon_e Z_e$. The magnitudes are independent because the original entries are independent. Signs can be chosen independent of all magnitudes even at zero atoms: the product law on (ε, Z) has exactly the required image law for Y . The sign at (i, j) is distinct from that at (j, i) when both labels exist. Both directions of the single edge between r_i and c_j use the same variable Y_{ij} .

4.1 The space and its polynomial domain

Let μ_Z be the joint magnitude law and define

$$\mathcal{H} = L_2(\mu_Z; \ell_2(\mathcal{V} \times 2^{\mathcal{E}})), \quad \|f\|_{\mathcal{H}}^2 = \mathbb{E}_Z \sum_{v \in \mathcal{V}} \sum_{S \subset \mathcal{E}} |f_{v,S}(Z)|^2. \quad (15)$$

Conditional Fourier transform in the signs is the isometry

$$(\mathcal{U}f)_v(\varepsilon, Z) = \sum_{S \subset \mathcal{E}} \varepsilon_S f_{v,S}(Z) \quad (16)$$

onto $L_2(\varepsilon, Z; \ell_2(\mathcal{V}))$. This transform preserves the L_2 norm, not general L_p norms. For example, with one sign and unit magnitude, amplitudes $f_{\emptyset} = f_{\{e\}} = 1$ have coefficient-vector norm $\sqrt{2}$, whereas $\mathcal{U}f = 1 + \varepsilon_e$ has L_4 norm $2^{3/4}$. Every non- L_2 norm used in the contraction below is therefore a norm of the reconstructed random polynomial $\mathcal{U}f$, rather than of its occupation-amplitude vector. All formulas below may be restricted to polynomial coefficients, for which every required moment is finite. No assertion about boundedness of the full multiplication operator on $\mathcal{H}$ is needed.

Define $\mathcal{P}_d^+$ to be the image under $\mathcal{U}^{-1}$ of vector polynomials in the original Y_e of degree at most d whose ordinary coefficients are nonnegative in each current-vertex coordinate. Thus a monomial $a_{v,\nu} Y^\nu$ contributes $a_{v,\nu} Z^\nu$ to the component (v, S) with $S = \{e : \nu_e \text{ is odd}\}$. This parity compatibility is part of the definition. It ensures that $\mathcal{U}f$ is a polynomial in the original variables, so Lemma 3.2 applies.

The parity grade of (v, S) is $|S|$. Let P_k keep exactly grade k . For $f \in \mathcal{P}_d^+$, necessarily $P_k f = 0$ for $k > d$. Projection onto any collection of pairs (v, S) simply discards ordinary monomials, hence preserves $\mathcal{P}_d^+$. Distinct such collections are orthogonal in (15). Magnitude coefficients on different parity states may have common even powers, including powers of edges outside the occupied set S . They are not required to be orthogonal or independent.

4.2 Multiplication, creation, and annihilation

The actual symmetric dilation is

$$H_Y = \begin{pmatrix} 0 & Y \\ Y^\top & 0 \end{pmatrix}. \quad (17)$$

For a scalar magnitude polynomial $h(Z)$ at state (v, S) , conjugating multiplication by H_Y through $\mathcal{U}$ gives

$$\mathcal{T}(h(Z)|v, S\rangle) = \sum_{e=vw} Z_e h(Z) |w, S \triangle \{e\}\rangle. \quad (18)$$

Here $e = vw$ means the original bipartite entry with endpoints v, w . Define $\mathcal{C}_+$ by keeping terms with $e \notin S$ and $\mathcal{C}_-$ by keeping terms with $e \in S$. Their sum is exactly $\mathcal{T}$, and they change parity grade by $+1$ and -1 .

Both maps multiply the magnitude coefficient by Z_e . On an ordinary monomial Y^ν , either action increases ν_e by one. Therefore

$$\mathcal{C}_+\mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+, \quad \mathcal{C}_-\mathcal{P}_d^+ \subset \mathcal{P}_{d+1}^+, \quad \mathcal{TP}_d^+ \subset \mathcal{P}_{d+1}^+. \quad (19)$$

In particular annihilation does not divide by a magnitude or reset its earlier powers. On Y_e^3 , its action gives Y_e^4 while removing the odd parity flag. This retains the full fourth moment of that same entry.

5 Exact localization by the two original count vectors

For $S \subset \mathcal{E}$, define

$$r_i(S) = |S \cap (\{i\} \times [n])|, \quad c_j(S) = |S \cap ([n] \times \{j\})|.$$

The vectors $r(S), c(S)$ count occupied original parity labels, not total magnitude degrees. Write δ_i for a coordinate vector in the relevant row or column count space.

Lemma 5.1 (Two exact count decompositions). *Consider maps from a current row to a current column, at input parity grade k . Each of the following gives orthogonal partitions of both the input states and the output states of the specified map.*

1. *For creation, label input (r_i, S) by $d = r(S) + \delta_i$ and output (c_j, U) by $d = r(U)$. Put $I = \{i : d_i > 0\}$. Then $|I| \leq k + 1$ and the entire sector is an input/output parity-count compression of multiplication by $Y[I, :]^\top$.*
2. *For annihilation, label input (r_i, S) by $d = c(S)$ and output (c_j, U) by $d = c(U) + \delta_j$. Put $J = \{j : d_j > 0\}$. Then $|J| \leq k$ and the entire sector is an input/output parity-count compression of multiplication by $Y[:, J]^\top$. At $k = 0$ the map is zero.*

The reverse current-side direction transposes the statements. Every magnitude variable remains available inside the coefficients.

Proof. Creation along $e = (i, j) \notin S$ produces $U = S \cup \{(i, j)\}$, so $r(U) = r(S) + \delta_i = d$. For fixed d , every input current row lies in I , and all parity flags on input and output lie in $I \times [n]$. The sum of the coordinates of d is $k + 1$, giving $|I| \leq k + 1$. Multiplication by $Y[I, :]^\top$ therefore includes every actual transition, with exactly its original coefficient.

More explicitly, let P_d select the input label $r(S) + \delta_i = d$ and Q_d select output grade $k + 1$ with $r(U) = d$. The sector map is $Q_d \mathcal{U}^{-1} M_{Y[I, :]}^\top \mathcal{U} P_d$, with the natural current-coordinate inclusions understood. Output grade removes every annihilation term. Conversely every selected creation term satisfies the same original input and output conditions. This proves equality of maps, rather than a comparison of their coefficients.

For annihilation, $e = (i, j) \in S$ produces $U = S \setminus \{(i, j)\}$, so $c(U) + \delta_j = c(S) = d$. Every parity flag lies in $[n] \times J$ and every output current column lies in J , although the input current row may be arbitrary. Now $\sum_j d_j = k$. Select input label $c(S) = d$ and output grade $k - 1$ with $c(U) + \delta_j = d$. The analogous compression of $M_{Y[:, J]}^\top$ retains exactly the annihilation terms. Creation terms have grade $k + 1$ and cannot survive the output projection.

Each input label is a function of (v, S) alone, and so is each output label. Distinct labels thus give disjoint components of (15), hence orthogonal decompositions. The two current-side directions have orthogonal domains and ranges. Different input grades also have different output grades within each of $\mathcal{C}_+$ and $\mathcal{C}_-$. No amplitude restriction was used: even powers outside the selected row or column set remain in $f_{v,S}(Z)$ throughout. $\square$

The distinction between the two labels is necessary. The creator is controlled through a small set of original rows in its output counts; the annihilator is controlled through a small set of original columns in its input counts. We will estimate the latter directly, not infer it by applying an adjoint argument to a cone-restricted norm inequality.

6 Thin matrices and the cone contraction

Lemma 6.1 (A net only on the small axis). *For a fixed original row set I , a fixed original column set J , and every $p \geq 1$,*

$$(\mathbb{E}\|Y[I, :]\|^p)^{1/p} \leq 2 \cdot 5^{|I|/p} C_{\text{SW}}(\alpha) \{\sigma_r(Y) + R_Y(p)\}, \quad (20)$$

$$(\mathbb{E}\|Y[:, J]\|^p)^{1/p} \leq 2 \cdot 5^{|J|/p} C_{\text{SW}}(\alpha) \{\sigma_c(Y) + R_Y(p)\}. \quad (21)$$

Empty submatrices give zero and require no net.

Proof. Fix a deterministic unit vector $s \in \mathbb{R}^I$. Independence and zero means give

$$\mathbb{E}\|Y[I, :]^\top s\|_2^2 = \sum_{i \in I} s_i^2 \sum_j \mathbb{E}Y_{ij}^2 \leq \sigma_r(Y)^2. \quad (22)$$

Its expected norm is at most $\sigma_r(Y)$. The weak p moment of this Hilbert vector is at most $R_Y(p)$, since every deterministic unit dual vector is a column vector t and the resulting scalar is $\sum_{i \in I, j} s_i t_j Y_{ij}$. Theorem 2.1 therefore bounds its strong p moment by $C_{\text{SW}}(\alpha)(\sigma_r(Y) + R_Y(p))$.

Choose a deterministic $1/2$ -net $\mathcal{N}$ of the unit sphere in $\mathbb{R}^I$, with $|\mathcal{N}| \leq 5^{|I|}$. This cardinality follows by taking a maximal $1/2$ -separated set and comparing its disjoint balls of radius $1/4$ with the ball of radius $5/4$. For every matrix realization, $\|Y[I, :]\| \leq 2 \max_{s \in \mathcal{N}} \|Y[I, :]^\top s\|_2$. Raising to p and bounding a maximum by a sum gives

$$(\mathbb{E}\|Y[I, :]\|^p)^{1/p} \leq 2 \left(\sum_{s \in \mathcal{N}} \mathbb{E}\|Y[I, :]^\top s\|_2^p \right)^{1/p}.$$

This proves (20). Transposing proves (21), using the maximum column variance. $\square$

Proposition 6.2 (Contraction on the preserved cone). *For an integer $q \geq 1$, put*

$$B_\alpha = 2 \cdot 5^{1/4} C_{\text{SW}}(\alpha) H_\alpha = 2\sqrt{3} \cdot 5^{1/4} \alpha C_{\text{SW}}(\alpha). \quad (23)$$

For every $f \in \mathcal{P}_{q-1}^+$,

$$\begin{aligned} \|\mathcal{C}_\pm f\|_{\mathcal{H}} &\leq B_\alpha \{\sigma(Y) + R_Y(4q)\} \|f\|_{\mathcal{H}}, \\ \|\mathcal{T}f\|_{\mathcal{H}} &\leq 2B_\alpha \{\sigma(Y) + R_Y(4q)\} \|f\|_{\mathcal{H}}. \end{aligned} \quad (24)$$

Proof. Set $p = 4q$. First consider row-to-column creation at a fixed parity grade $k \leq q - 1$ and decompose its input as $\sum_d f_d$ using Lemma 5.1. Every f_d lies in $\mathcal{P}_{q-1}^+$, since its projection only discards ordinary monomials. Its sector has $|I| \leq k + 1 \leq q$. Apply $\mathcal{U}$ first. It identifies the multiplier in that sector with $Y[I, :]^\top$, and the transformed output projection is an L_2 contraction. Hölder is then applied on the original joint probability space to $Y[I, :]^\top \mathcal{U}f_d$. Together, Lemma 3.2 and Lemma 6.1 imply

$$\begin{aligned} \|\mathcal{C}_{+,d} f_d\|_{\mathcal{H}} &\leq \|Y[I, :]^\top \mathcal{U}f_d\|_{L_2(\ell_2)} \\ &\leq (\mathbb{E}\|Y[I, :]\|^p)^{1/p} \|\mathcal{U}f_d\|_{L_{2p/(p-2)}(\ell_2)} \\ &\leq 2 \cdot 5^{|I|/p} C_{\text{SW}}(\alpha) \{\sigma_r(Y) + R_Y(p)\} H_\alpha^{4(q-1)/p} \|f_d\|_{\mathcal{H}} \end{aligned}$$

$$\leq B_\alpha\{\sigma_r(Y) + R_Y(4q)\}\|f_d\|_{\mathcal{H}}. \quad (25)$$

The expectation in Hölder is over the full joint sign and magnitude law. It requires no independence between the matrix multiplier and $\mathcal{U}f_d$, so magnitude powers outside I stay in the state. The $L_{2p/(p-2)}$ factor is explicitly the norm of $\mathcal{U}f_d$. Only its final L_2 norm is replaced by $\|f_d\|_{\mathcal{H}}$ using the isometry; no L_p isometry of occupation amplitudes is used.

For row-to-column annihilation use its column-count sectors instead. They have $|J| \leq k \leq q-1$, and (21) gives the same bound with $\sigma_c(Y)$. The reverse direction transposes these two statements. Sum the squared estimates over the orthogonal input and output sectors, over their grades, and over current-side directions. The sums of squared input norms equal $\|f\|_{\mathcal{H}}^2$; there is no factor for the number of sectors. This proves the separate bounds for $\mathcal{C}_+$ and $\mathcal{C}_-$. Their triangle inequality gives the stated bound for $\mathcal{T}$. $\square$

There is no random maximum over row or column sets in this argument. Each set is the support of a deterministic count label, and the same unconditional inequality bounds every corresponding squared sector norm before summation. Nor is (24) a bound on the entire polynomial linear span: it is asserted only on the specified cone at the specified total degree.

7 Vacuum walks and the local moment theorem

Theorem 7.1 (Symmetric regular-law local moment bound). *Let Y have independent symmetric entries satisfying (1). With B_α from (23), every integer $q \geq 1$ satisfies*

$$(\mathbb{E}\|Y\|^{2q})^{1/(2q)} \leq 2B_\alpha n^{1/(2q)}\{\sigma(Y) + R_Y(4q)\}. \quad (26)$$

Proof. For a current vertex $v \in \mathcal{V}$, let Ω_v be the state with coefficient one at $(v, \emptyset)$ and zero elsewhere. It has norm one in $\mathcal{H}$. Under $\mathcal{U}$ it is the constant vector e_v . For every $j \geq 0$,

$$\mathcal{U}\mathcal{T}^j\Omega_v = H_Y^j e_v.$$

Each coordinate on the right is the sum, over original ordered walks of length j between the indicated vertices, of the product of their entry variables. Every coefficient is a nonnegative integer, even when the realized entries are negative. Repeated edges retain their powers in the same monomial. Thus $\mathcal{T}^j\Omega_v \in \mathcal{P}_j^+$.

Apply Proposition 6.2 successively to the prefixes $j = 0, \dots, q-1$, keeping the common horizon q :

$$\|\mathcal{T}^q\Omega_v\|_{\mathcal{H}} \leq [2B_\alpha\{\sigma(Y) + R_Y(4q)\}]^q. \quad (27)$$

Since H_Y is real symmetric and its two diagonal blocks in the even power have equal traces,

$$\begin{aligned} 2\mathbb{E} \operatorname{tr}(YY^\top)^q &= \mathbb{E} \operatorname{tr} H_Y^{2q} = \sum_{v \in \mathcal{V}} \mathbb{E} \|H_Y^q e_v\|_2^2 \\ &= \sum_{v \in \mathcal{V}} \|\mathcal{T}^q\Omega_v\|_{\mathcal{H}}^2 \leq 2n[2B_\alpha\{\sigma(Y) + R_Y(4q)\}]^{2q}. \end{aligned} \quad (28)$$

Finally $\|Y\|^{2q} \leq \operatorname{tr}(YY^\top)^q$. Divide by two and take the $2q$ -th root to obtain (26). $\square$

The trace normalization is the original dimension $2n$, rather than the size of the parity state space. Every equality uses the original joint law. All expectations are finite by the entry moment assumptions. The proof requires neither polynomial completeness in an infinite L_2 space nor a bounded extension of the full multiplication operator. At q comparable to $\log n$, the factor $n^{1/(2q)}$ is absolute. The following reductions supply the correct local order and the exact shared-index deletion statement, including the nonsymmetric case.

8 From the local estimate to shared-index deletion

We now give the matrix-level reduction, including the deterministic permutation and the estimate of its remainder. The construction follows Latała–Świątkowski [4, Remark 4.5]; its underlying variance-envelope estimate is a consequence of Latała–van Handel–Youssef [5, Theorem 4.4]. We state that additional established input explicitly below and prove the specialization needed here. Thus the deletion implication is not being used as an unexplained matrix theorem.

Throughout this section constants denoted by C_α may change between occurrences and depend only on the regularity constant. Write $p_* = \log 2$ and $\beta = \max\{1/2, \log_2 \alpha\}$.

8.1 Scalar comparison, including the subunit endpoint

The scalar moment comparison supplied by [2, Corollary 1.4] says that there is $K_\alpha \geq 1$ such that every deterministic linear combination S of independent centered α -regular variables satisfies

$$\|S\|_{L_p} \leq K_\alpha (p/r)^\beta \|S\|_{L_r}, \quad p \geq r \geq 2. \quad (29)$$

In particular there is $J_\alpha \geq 1$ for which

$$\|S\|_{L_{2p}} \leq J_\alpha \|S\|_{L_p}, \quad p \geq p_*. \quad (30)$$

Here is the small-order justification, so that no norm inequality below order one is implicit. Set $v = \mathbb{E}S^2$ and $K_4 = \max\{3, \alpha^4\}$. If $S = \sum_e a_e X_e$, independence and zero means give

$$\mathbb{E}S^4 = \sum_e a_e^4 \mathbb{E}X_e^4 + 6 \sum_{e < f} a_e^2 a_f^2 \mathbb{E}X_e^2 \mathbb{E}X_f^2 \leq K_4 v^2.$$

For $v > 0$, the classical Paley–Zygmund inequality [10] applied to S^2 yields

$$\mathbb{P}\{|S| \geq \sqrt{v/2}\} \geq (4K_4)^{-1}, \quad \|S\|_{L_p} \geq 2^{-1/2} (4K_4)^{-1/p_*} \sqrt{v} \quad (p \geq p_*).$$

For $p \in [p_*, 2]$, monotonicity gives $\|S\|_{L_{2p}} \leq \|S\|_{L_4} \leq K_4^{1/4} \sqrt{v}$. For $p \geq 2$, use (29). Consequently (30) holds, for example, with

$$J_\alpha = \max\{1, K_\alpha 2^\beta, \sqrt{2} K_4^{1/4} (4K_4)^{1/p_*}\}.$$

When $v = 0$, all the assertions are immediate. Taking bilinear suprema on a fixed original submatrix gives

$$R_{X,I}(2p) \leq J_\alpha R_{X,I}(p), \quad p \geq p_*. \quad (31)$$

The set I stays fixed during this operation.

8.2 The local estimate in the matrix-symmetric convention

For independent symmetric-in-law entries Y , Theorem 7.1 and (26), with $q = \lceil \log(n+1) \rceil$ for $n \geq 2$, imply

$$\mathbb{E}\|Y\| \leq L_\alpha \{\sigma(Y) + R_Y(\log(n+1))\}. \quad (32)$$

Indeed $4q \leq 8 \log(n+1)$, $n^{1/(2q)} \leq \sqrt{e}$, and three applications of (31) compare the two weak orders. The case $n = 1$ follows from $\mathbb{E}|Y_{11}| \leq \|Y_{11}\|_2$. All constants in the local theorem, including the improved positive-cone constant, are absorbed in L_α .

Let now $A = A^\top$ be an $m \times m$ matrix whose upper-triangular entries, including the diagonal, are independent symmetric variables with the same regularity bound. A reflected lower-triangular entry is the same variable, not another independent copy. Define

$$a(A) = \max_i \left(\sum_j \mathbb{E} A_{ij}^2 \right)^{1/2}, \quad \mathcal{R}_A(p) = \sup_{\|s\|_2, \|t\|_2 \leq 1} \|s^\top A t\|_{L_p}.$$

Writing $A = U + U^\top + D_0$, where U is strictly upper triangular, produces a matrix U with independent entries and $\sigma(U) \leq a(A)$. For $p \geq 1$,

$$R_U(p) \leq \mathcal{R}_A(p). \quad (33)$$

To see this, fix s, t . Absorb the signs of $s_i t_j$, $i < j$, into the independent symmetric variables. Increase the resulting coefficients $|s_i t_j|$ to $|s_i| |t_j| + |s_j| |t_i|$ and include the diagonal coefficients $|s_i| |t_i|$. Conditional on all magnitudes, averaging independent signs and using convexity of $|\cdot|^p$ shows that the moment does not decrease. The enlarged sum is $|s|^\top A |t|$. This reasoning also proves that a deterministic symmetric zero-one entry mask decreases $\mathcal{R}_A(p)$ for $p \geq 1$.

For $m \geq 2$, put $p_m = \log(m+1) > 1$. The diagonal obeys

$$\mathbb{E} \|D_0\| \leq \left(\sum_i \mathbb{E} |A_{ii}|^{p_m} \right)^{1/p_m} \leq e \mathcal{R}_A(p_m).$$

Combining this with (32) and (33) gives

$$\mathbb{E} \|A\| \leq C_\alpha \{a(A) + \mathcal{R}_A(\log(m+1))\}. \quad (34)$$

For $m = 1$, the variance bound suffices. The same statement applies, with its own dimension, to every principal submatrix and to every symmetric zero-one mask of that submatrix. These are the only block matrices used below.

We also need their strong moments. The established independent-coordinate strong/weak theorem [2, Theorem 1.1], applied with the operator norm, yields

$$(\mathbb{E} \|B\|^p)^{1/p} \leq C_\alpha \{\mathbb{E} \|B\| + \mathcal{R}_B(p)\}, \quad p \geq 1. \quad (35)$$

This use is valid for independent upper-triangular coordinates: their deterministic coefficient matrices are $E_{ij} + E_{ji}$ and E_{ii} . For clarity, the weak functional of the operator-norm Banach space equals the bilinear weak functional when $p \geq 1$. The nuclear-norm unit ball is the convex hull of rank-one matrices $st^\top$ with $\|s\|_2, \|t\|_2 \leq 1$; a singular-value decomposition and Minkowski give one inequality, and rank-one tests give the other. No independence between reflected entries is asserted.

8.3 The established variance-envelope input and its specialization

We use only the following upper, operator-norm case of [5, Theorem 4.4]. Suppose $G_{ij} = b_{ij} h_{ij}$ is matrix-symmetric, $b_{ij} = b_{ji} \geq 0$, and the upper-triangular h_{ij} are independent centered variables such that, for fixed $c_1, c_2 > 0$ and $\beta \geq 1/2$,

$$c_1 p^\beta \leq \|h_{ij}\|_{L_p} \leq c_2 p^\beta \quad \text{for all } p \geq 2.$$

If b_i^* is the nonincreasing rearrangement of the row maxima $\max_j b_{ij}$, that theorem gives

$$\mathbb{E} \|G\| \leq C(c_1, c_2, \beta) \left\{ \max_i \left(\sum_j b_{ij}^2 \right)^{1/2} + \sup_i b_i^* \log(i+1)^\beta \right\}. \quad (36)$$

Replacing the source's $\log i$ by $\log(i+1)$ only enlarges its upper bound. This is an additional established matrix theorem; it is not derived here from the scalar or vector moment comparisons.

Here is the full specialization that will control the remainder. For a matrix-symmetric W with independent symmetric regular upper-triangular variables, in *any fixed ordering* of its indices,

$$\mathbb{E}\|W\| \leq C_\alpha \left\{ \max_i \left(\sum_j \mathbb{E}W_{ij}^2 \right)^{1/2} + \sup_i \log(i+1)^\beta \max_j \|W_{ij}\|_2 \right\}. \quad (37)$$

To prove this consequence, put $b_{ij} = \|W_{ij}\|_2$, omitting zero variables. Iterated regularity gives $\|W_{ij}/b_{ij}\|_{L_p} \leq \alpha p^\beta$ for $p \geq 2$. If a symmetric scalar V has this bound, Markov gives

$$\mathbb{P}\{|V| > e\alpha p^\beta\} \leq e^{-p}, \quad p \geq 2.$$

Let E_0 have the exponential distribution of mean one and let $h = \varepsilon(1 + E_0)^\beta$, with an independent symmetric sign. For $c = e\alpha 2^\beta$, the preceding tail bound implies stochastic domination $|V| \leq_{\text{st}} c|h|$. In fact the right-hand magnitude is at least c , and for $t \geq c$, setting $u = (t/(e\alpha))^{1/\beta} \geq 2$ gives

$$\mathbb{P}\{|V| > t\} \leq e^{-u} \leq e^{1-u/2} = \mathbb{P}\{c(1 + E_0)^\beta > t\}.$$

Couple the magnitudes separately for every unordered matrix entry using independent uniform variables and the inverse distribution functions. Reattach independent signs. Conditional sign contraction for the convex matrix norm then bounds $\mathbb{E}\|W\|$ by $c\mathbb{E}\|(b_{ij}h_{ij})\|$.

The auxiliary variables meet the two-sided hypothesis of (36), with constants depending only on β . The lower bound follows by restricting to $E_0 \geq p$: $\|(1+E_0)^\beta\|_{L_p} \geq (1+p)^\beta e^{-1}$. The upper bound follows from $\mathbb{E}(1+E_0)^{\beta p} = e \int_1^\infty t^{\beta p} e^{-t} dt$ and the elementary gamma-integral bound $\Gamma(u+1)^{1/u} \leq C \max\{1, u\}$ for $u \geq 1$. Finally, if $b_i = \max_j b_{ij}$ and $K = \sup_i b_i \log(i+1)^\beta$ in the given ordering, then $b_i^* \log(i+1)^\beta \leq K$: among any i indices attaining the i largest values, at least one index is at least i . This proves (37) with no requirement that the ordering used in the construction below sort all row maxima.

8.4 Nested deterministic deletions and screened coefficients

For a matrix-symmetric A as above, let $\gamma = D_{\text{sym}}(A)$ be the functional in (4), formed with $\mathcal{R}$ instead of R , and put $a = a(A)$. We prove

$$\mathbb{E}\|A\| \leq C_\alpha(a + \gamma). \quad (38)$$

Appending deterministic zero rows and columns changes neither a nor γ . At deletion budgets below the original dimension, any deleted zero index can be discarded; larger budgets give a zero minimum by deleting all original indices. We may therefore pad the matrix to dimension N_L , where $L \geq 2$,

$$N_0 = 1, \quad N_k = 2^{2^k} \ (k \geq 1), \quad M_k = N_k + N_k N_{k-1}.$$

Only finitely many padded indices will be used. Notice that $M_k + N_k \leq N_{k+1}$ and $M_k < N_{k+1}$ for every $k \geq 1$.

For $1 \leq k < L$, choose a minimizer F_k , of cardinality at most N_k , such that

$$\mathcal{R}_{A[F_k^c, F_k^c]}(\log(N_k + 1)) \leq \gamma.$$

Construct a permutation and nested initial sets I_k of exactly N_k indices for $1 \leq k \leq L$ as follows. Start with $I_1 \supset F_1$. Given I_k with $k < L$, select from its complement the N_{k-1} largest

standard deviations in each column indexed by I_k , using all remaining indices if a column has fewer candidates. Their union has size at most $N_k N_{k-1}$. Put this union immediately after I_k , fill as needed to position M_k , then include F_k and fill to position N_{k+1} . The displayed size inequality guarantees room for these choices. All choices use only the laws, and may be made deterministic by a fixed tie-breaking rule. In the resulting index order $I_k = [N_k]$, and, for $k \geq 2$,

$$\mathcal{R}_{A[I_k^c, I_k^c]}(\log(N_{k-1} + 1)) \leq \gamma. \quad (39)$$

This follows from $F_{k-1} \subset I_k$ and the symmetric contraction proved above. For I_1 , the same assertion holds at $\log(N_1 + 1)$.

The screening step has a second consequence:

$$j \leq N_k, \quad i > M_k \implies \|A_{ij}\|_2 \leq a N_{k-1}^{-1/2}. \quad (40)$$

Indeed, in the remaining part of column j , each unselected entry is no larger than the N_{k-1} selected ones. The sum of squared standard deviations in that column is at most a^2 . This argument does not replace an entry or identify two original variables.

8.5 Two block families and their maximum norms

Intersect every interval below with $[N_L]$ and set

$$J_1 = [1, M_1], \quad J_k = (N_{k-1}, M_k] \quad (2 \leq k \leq L).$$

Intervals with the same parity of k are disjoint, because $M_k < N_{k+1}$. Let

$$E_U = \bigcup_{k \text{ odd}} J_k \times J_k, \quad E_V = \left(\bigcup_{k \text{ even}} J_k \times J_k \right) \setminus E_U, \quad E_W = [N_L]^2 \setminus (E_U \cup E_V),$$

and write $A = U + V + W$ by these deterministic masks. Every mask is symmetric. Both U and V are block diagonal; their nonzero blocks B_k are either $A[J_k, J_k]$ or a symmetric zero-one mask of it. Their upper-triangular coordinates retain their original independence.

Set $p_k = 2^{k+1}$. For $k \geq 2$ let $h_k = \max\{1, k-2\}$. The block J_k lies outside I_{k-1} , its dimension is at most M_k , and elementary estimates on N_k, M_k give

$$\log(|J_k| + 1) \leq p_k, \quad p_k \leq 16 \log(N_{h_k} + 1).$$

For $k \geq 3$, (39) controls the complement of I_{k-1} at $\log(N_{k-2} + 1)$; for $k = 2$, the initial I_1 estimate supplies $\log(N_1 + 1)$. Thus symmetric contraction and four scalar doublings give

$$\mathcal{R}_{B_k}(p_k) \leq J_\alpha^4 \gamma, \quad k \geq 2.$$

The local mean estimate (34) and the strong-moment estimate (35) therefore yield

$$(\mathbb{E}\|B_k\|^{p_k})^{1/p_k} \leq C_\alpha(a + \gamma). \quad (41)$$

The first block has dimension at most $M_1 = 8$. Its fourth moment is bounded directly by the Frobenius norm and Minkowski:

$$(\mathbb{E}\|B_1\|^4)^{1/4} \leq \left(\sum_{i,j \in J_1} \|A_{ij}\|_4^2 \right)^{1/2} \leq \alpha \sqrt{8} a.$$

Thus (41) holds for all blocks.

The passage from block moments to the expectation of their maximum does not cost the number of blocks. If $Z_k \geq 0$ and $\|Z_k\|_{L_{p_k}} \leq K$ with $p_k = 2^{k+1}$, then for $t \geq 2$,

$$\mathbb{P}\{\max_k Z_k > tK\} \leq \sum_{k \geq 1} t^{-2^{k+1}} \leq \sum_{k \geq 1} t^{-k-3} \leq 2t^{-4}.$$

Integration gives $\mathbb{E} \max_k Z_k \leq 3K$. Independence is unnecessary for this step. Consequently

$$\mathbb{E}\|U\| + \mathbb{E}\|V\| \leq C_\alpha(a + \gamma). \quad (42)$$

8.6 The remaining entries have a summable variance envelope

For $M_k < i \leq M_{k+1}$, $k \geq 1$, a nonzero entry of W satisfies

$$\max_j \|W_{ij}\|_2 \leq aN_{k-1}^{-1/2}. \quad (43)$$

To check all index cases, first suppose $i \leq N_{k+1}$. Then $i \in J_{k+1}$, so a surviving j must satisfy either $j \leq N_k$ or $j > M_{k+1}$. The first case uses (40) at k ; the second uses it at $k+1$ and symmetry. If $N_{k+1} < i \leq M_{k+1}$, then i belongs to both J_{k+1} and J_{k+2} whenever the latter interval is present. A surviving j has $j \leq N_k$ or $j > M_{k+2}$, and screening at k or $k+2$ gives the claimed bound. At the final padded endpoint the second alternative is empty. For $i \leq M_1$, the trivial bound $\max_j \|W_{ij}\|_2 \leq a$ suffices.

Masking can only decrease row variances. Moreover

$$\sup_i \log(i+1)^\beta \max_j \|W_{ij}\|_2 \leq a \max \left\{ (\log 9)^\beta, \sup_{k \geq 1} \frac{\log(M_{k+1}+1)^\beta}{\sqrt{N_{k-1}}} \right\} \leq C_\alpha a.$$

The last supremum is finite: its terms for $k \geq 2$ are at most $C_\beta 2^{(k+1)\beta} 2^{-2^{k-2}}$. Applying the proved specialization (37) gives $\mathbb{E}\|W\| \leq C_\alpha a$. Together with (42), this proves (38). Every subset and mask used in this argument is deterministic and symmetric.

8.7 The actual dilation and its paired deletion budget

Return to an $n \times n$ matrix Y with independent symmetric entries and use its actual dilation

$$H_Y = \begin{pmatrix} 0 & Y \\ Y^\top & 0 \end{pmatrix}.$$

Its independent upper-triangular variables are exactly the original Y_{ij} , its norm is $\|Y\|$, and $a(H_Y) = \sigma(Y)$. For any random rectangular matrix Z , even with dependent entries,

$$\mathcal{R}_{H_Z}(p) = R_Z(p), \quad p \geq 1. \quad (44)$$

The lower inequality takes the two test vectors in opposite halves. For the upper one, decompose $s = (s_r, s_c)$ and $t = (t_r, t_c)$; Minkowski bounds the two terms by

$$R_Z(p) \{ \|s_r\|_2 \|t_c\|_2 + \|s_c\|_2 \|t_r\|_2 \} \leq R_Z(p) \|s\|_2 \|t\|_2.$$

This equality is not invoked at subunit orders.

For a dilation deletion budget $2 \leq k \leq 2n$, put $\ell = \lfloor k/2 \rfloor$. Choose an original minimizer I with $|I| \leq \ell$ at order $\log(\ell+1)$, and delete both copies $I \cup (n+I)$. This uses at most $2\ell \leq k$ dilation indices, and the remainder is exactly $H_{Y_{[I^c, I^c]}}$. Since $\log(k+1) \leq 2\log(\ell+1)$ and $\log(k+1) > 1$, equations (31) and (44) show that its weak moment is at most $J_\alpha D(Y)$. This includes $\ell = n$, when it is zero. For $k = 1$, choose no deletion and use instead

$$\mathcal{R}_{H_Y}(\log 2) \leq \mathcal{R}_{H_Y}(2) = R_Y(2) = \max_{ij} \|Y_{ij}\|_2 \leq \sigma(Y).$$

Here the middle equality for $R_Y(2)$ follows from independence and zero means: its squared value is the supremum of $\sum_{ij} \mathbb{E} Y_{ij}^2 s_i^2 t_j^2$, attained at a coordinate pair. We conclude that

$$D_{\text{sym}}(H_Y) \leq \max\{\sigma(Y), J_\alpha D(Y)\}, \quad \mathbb{E}\|Y\| \leq C_\alpha \{M(Y) + D(Y)\}. \quad (45)$$

The intermediate deletion of two copies has been paid for explicitly; $D(Y)$ still deletes one shared original set from both axes.

8.8 Symmetrization without changing the small exponent

For general independent centered entries X , let X' be an independent copy and put $Y = X - X'$. For every $p \geq 1$, conditional Jensen and the triangle inequality give

$$\|X_{ij}\|_{L_p} \leq \|Y_{ij}\|_{L_p}, \quad \|Y_{ij}\|_{L_{2p}} \leq 2\alpha\|Y_{ij}\|_{L_p}.$$

Thus Y has independent symmetric 2α -regular entries, $M(Y) = \sqrt{2}M(X)$, and $\mathbb{E}\|X\| \leq \mathbb{E}\|Y\|$. For each fixed I, s, t , write its bilinear sum as S and the independent copy as S' . If $p \geq 1$, then $\|S - S'\|_{L_p} \leq 2\|S\|_{L_p}$. If $p_* \leq p < 1$, subadditivity of the p th power gives instead

$$\|S - S'\|_{L_p} \leq 2^{1/p}\|S\|_{L_p} \leq e\|S\|_{L_p}.$$

Taking the same bilinear suprema, deterministic minima, and budget maximum therefore gives $D(Y) \leq eD(X)$. Applying (45) with parameter 2α proves

$$\mathbb{E}\|X\| \leq C_{2\alpha}\{\sqrt{2}M(X) + eD(X)\} \leq C'_\alpha\{M(X) + D(X)\}. \quad (46)$$

9 The known lower bound on the same matrix

For completeness we reproduce the argument of [4, Theorem 4.1 and Lemma 4.2], with its small-order and independence details. In this section X itself has independent centered entries satisfying (1). There is no symmetrization or comparison to a second input matrix.

9.1 Row and column variances

Fix a row and put $Q = \sum_j X_{ij}^2$, $v = \mathbb{E}Q$. Regularity at order two and Cauchy–Schwarz give

$$\mathbb{E}Q^2 = \sum_{j,l} \mathbb{E}(X_{ij}^2 X_{il}^2) \leq \alpha^4 \sum_{j,l} \mathbb{E}X_{ij}^2 \mathbb{E}X_{il}^2 = \alpha^4 v^2.$$

Paley–Zygmund implies $\mathbb{E}\sqrt{Q} \geq (4\sqrt{2}\alpha^4)^{-1}\sqrt{v}$. Since every row and column Euclidean norm is at most $\|X\|$,

$$M(X) \leq 8\sqrt{2}\alpha^4 \mathbb{E}\|X\|. \quad (47)$$

Rows or columns with zero variance cause no exception.

9.2 A large weak moment has a small original support

For $p \geq 2$, let $R_X^{\text{sp}}(p)$ be the bilinear supremum restricted to vectors whose supports each have cardinality at most $p^{2\beta}$. Then

$$R_X(p) \leq R_X^{\text{sp}}(p) + C_\alpha M(X). \quad (48)$$

Here is a direct proof. For a unit vector z , set $I_p(z) = \{i : |z_i| \geq p^{-\beta}\}$, so that $|I_p(z)| \leq p^{2\beta}$. Split a bilinear sum into the part on $I_p(s) \times I_p(t)$, the part with $j \notin I_p(t)$, and the remaining part with $i \notin I_p(s)$ and $j \in I_p(t)$. The first part is controlled by $R_X^{\text{sp}}(p)$. For the second, independence and zero means give

$$\left\| \sum_{i, j \notin I_p(t)} s_i t_j X_{ij} \right\|_2^2 \leq p^{-2\beta} \sum_i s_i^2 \sum_j \mathbb{E}X_{ij}^2 \leq p^{-2\beta} \sigma_r(X)^2.$$

Apply (29) with lower order two to bound its L_p moment by $C_\alpha \sigma_r(X)$. The third part is bounded in the same way by $C_\alpha \sigma_c(X)$. The triangle inequality proves (48), also on every fixed deleted submatrix with the same constant. The union of the two sparse vector supports uses at most $2p^{2\beta}$ original indices.

9.3 A uniform scalar lower tail

Choose $D_\alpha \geq e$ so large that $\|S\|_{L_{2r}} \leq D_\alpha \|S\|_{L_r}$ for $r \geq 2$, using (29), and set $\delta = (4 \log D_\alpha)^{-1} \leq 1/4$. For $P \geq 2/\delta$, take $r = \delta P \geq 2$. Paley-Zygmund applied to $|S|^r$ and scalar moment comparison give

$$\begin{aligned} \mathbb{P}\{|S| \geq \tfrac{1}{2}\|S\|_{L_r}\} &\geq (1 - 2^{-r})^2 \left(\frac{\|S\|_{L_r}}{\|S\|_{L_{2r}}} \right)^{2r} \geq \tfrac{1}{4}e^{-P/2}, \\ \|S\|_{L_r} &\geq K_\alpha^{-1} \delta^\beta \|S\|_{L_P}. \end{aligned}$$

Thus, with $c_0 = \delta^\beta / (2K_\alpha) > 0$,

$$\mathbb{P}\{|S| \geq c_0 \|S\|_{L_P}\} \geq \tfrac{1}{4}e^{-P/2}. \quad (49)$$

Only the scalar comparison of the original independent entries is used.

9.4 Disjoint witnesses for each deletion budget

Fix $1 \leq k \leq n$ and write

$$P = \log(k+1), \quad \gamma_k = \min_{|I| \leq k} R_{X,I}(P).$$

We prove $\gamma_k \leq C_\alpha \mathbb{E}\|X\|$ uniformly in k . Choose $k_0(\alpha)$ large enough that, for $k \geq k_0$,

$$P \geq 2/\delta, \quad 2P^{2\beta} \leq \sqrt{k}, \quad k \geq 4.$$

For $k < k_0$, moment monotonicity and (29) show

$$\gamma_k \leq R_X(P) \leq C_\alpha \max_{ij} \|X_{ij}\|_2 \leq C_\alpha M(X).$$

For $P < 2$, only monotonicity to order two is used, so this covers $k = 1$ with its unchanged exponent $\log 2$. Equation (47) completes these finitely many orders.

Now let $k \geq k_0$. If $\gamma_k \leq 2C_\alpha M(X)$ with the constant from (48), the variance bound again suffices. Otherwise, for every deterministic $|I| \leq k$, sparsification in $X[I^c, I^c]$ supplies unit vectors s, t supported outside I on a common set J of at most $2P^{2\beta} \leq \sqrt{k}$ indices, with

$$\left\| \sum_{i,j \in J} s_i t_j X_{ij} \right\|_{L_P} \geq \gamma_k/2.$$

The suprema are attained: there are finitely many possible supports, and the moment functional is continuous on each compact product of unit balls. Alternatively an arbitrarily small slack gives the same constants.

Apply this observation successively, deleting the union of the previously selected sets. It constructs $m = \lfloor \sqrt{k} \rfloor$ disjoint deterministic sets $J_1, \dots, J_m$ and bilinear sums $S_1, \dots, S_m$ with $\|S_l\|_{L_P} \geq \gamma_k/2$. At every stage the already deleted union has at most $(m-1)\sqrt{k} \leq k$ elements. The random variables S_l are independent because they use disjoint collections of original entries. Each satisfies $|S_l| \leq \|X\|$ pointwise, and (49) gives

$$\mathbb{P}\{|S_l| \geq c_0 \gamma_k/2\} \geq \frac{1}{4\sqrt{k+1}}.$$

Consequently

$$\begin{aligned} \mathbb{E}\|X\| &\geq \frac{c_0 \gamma_k}{2} \left[1 - \left(1 - \frac{1}{4\sqrt{k+1}} \right)^m \right] \\ &\geq \frac{c_0 \gamma_k}{2} \left(1 - e^{-1/(8\sqrt{2})} \right). \end{aligned}$$

In the last step we used $m \geq \sqrt{k}/2$ and $k \geq 4$. Taking the maximum over the original budgets proves

$$D(X) \leq C_\alpha \mathbb{E} \|X\|. \quad (50)$$

Combining (47) and (50) gives the known same-matrix lower estimate. Together with (46), it proves Theorem 1.1. For $n = 1$, $D(X) = 0$ and the direct comparison $\|X_{11}\|_2 \leq \alpha \mathbb{E} |X_{11}|$ also verifies the lower endpoint. All deletion minima throughout this proof remove the same deterministic original set from both axes.

10 Scope and the information retained by the proof

The estimate concerns the expected norm of the same original matrix throughout. Its constants depend only on the entry regularity parameter. The deletion sets are deterministic for the law, use the same original indices on both axes, and retain the exact moment order $\log(k+1)$. The lower comparison and the deletion construction have their stated public origins; the local argument developed here is the parity-count contraction on positive original-coefficient walk polynomials.

Three features of that local argument prevent unsupported reductions. First, the magnitudes and their repeated powers are never replaced by static entry heights or fresh copies. Second, sector orthogonality comes from the original signs and current vertices, not from magnitude monomials on disjoint rows. Third, the multiplier estimate is used only on a cone preserved by the actual walk prefixes. It does not bound all signed linear combinations in a polynomial span. The Hilbert Boolean estimate and the elementary scalar product-moment comparison were proved explicitly; the independent-coordinate strong/weak theorem and the variance-envelope input to deletion were stated with their hypotheses and citations.

This manuscript is a research proof draft of the stated mean-norm theorem. Its internal reviews and finite exact checks are supporting evidence, not by themselves formal verification.

Separate formal upper comparison. The MI-32 catalogue reports a checked proof of `MI32.main_upper` at the pinned source in [1, 13]. Although its statement existentially quantifies a constant, the proof supplies the explicit Lean term

```
SymmetricDeletion.deletionConstant (2 * α) * Real.exp 1.
```

The record concerns the upper comparison, not the lower bound reproduced above. It also records deliberate proof differences: the formal far-remainder argument avoids the external variance-envelope theorem, and its deletion decomposition acts on the original rectangular matrix rather than its symmetric dilation. Thus this record is not a line-by-line certification of the analytic proof printed here. The attribution revision does not replay Lean or identify later analytic constants with the formal term.

Limitations. No efficient algorithm for evaluating $D(X)$ or $R_X(p)$ is claimed. The moment estimates can supply upper-tail bounds by Markov’s inequality, but do not by themselves identify matching quantiles. Likewise, arbitrary noncommuting coefficient insertions can destroy the nonnegative monomial cone; applications with such insertions need their own preserved-coefficient argument.

References

- [1] *MI-32: Spectral norms of independent entries with regular moment growth*. Problem statement, primary-source references, and reported verification record; consulted 16 September 2026. <https://github.com/ajt60gaibb/OpenProblemsInNLA/blob/main/matrix-inequalities-and-norms/MI-32/README.md>.
- [2] R. Latała and M. Strzelecka. *Comparison of weak and strong moments for vectors with independent coordinates*. *Mathematika* **64** (2018), 211–229. <https://arxiv.org/abs/1612.02407>.
- [3] R. O’Donnell. *Analysis of Boolean Functions*, Lecture 16, Carnegie Mellon University. Boolean hypercontractivity, Theorem 1.1. <https://www.cs.cmu.edu/~odonnell/boolean-analysis/lecture16.pdf>.
- [4] R. Latała and W. Świątkowski. *Norms of Randomized Circulant Matrices*. *Electronic Journal of Probability* **27** (2022), article 80, 1–23. The cited theorem and remark numbers refer to arXiv version 2. <https://arxiv.org/abs/2106.03139v2>.
- [5] R. Latała, R. van Handel, and P. Youssef. *The dimension-free structure of nonhomogeneous random matrices*. *Inventiones Mathematicae* **214** (2018), 1031–1080. <https://arxiv.org/abs/1711.00807v2>.
- [6] A. Bonami. *Étude des coefficients de Fourier des fonctions de $L^p(G)$* . *Annales de l’Institut Fourier* **20**(2) (1970), 335–402. doi:10.5802/aif.357.
- [7] W. Beckner. *Inequalities in Fourier analysis*. *Annals of Mathematics* **102**(1) (1975), 159–182. doi:10.2307/1970980.
- [8] R. O’Donnell. *Analysis of Boolean Functions*. Cambridge University Press, 2014. Author’s freely available edition: <https://arxiv.org/abs/2105.10386>.
- [9] L. Accardi and M. Bożejko. *Interacting Fock spaces and Gaussianization of probability measures*. *Infinite Dimensional Analysis, Quantum Probability and Related Topics* **1**(4) (1998), 663–670. doi:10.1142/S0219025798000363.
- [10] R. E. A. C. Paley and A. Zygmund. *On some series of functions, (3)*. *Proceedings of the Cambridge Philosophical Society* **28**(2) (1932), 190–205. doi:10.1017/S0305004100010860.
- [11] R. Latała. *On the spectral norm of Rademacher matrices*. Preprint, 2024; arXiv:2405.13656v2, revised 18 August 2025. The cited theorem numbers refer to version 2. <https://arxiv.org/abs/2405.13656v2>.
- [12] R. Meller. *Spectral norm of matrices with independent entries up to polyloglog*. Preprint, 2025; arXiv:2512.23673v2, revised 29 January 2026. <https://arxiv.org/abs/2512.23673v2>.
- [13] D. Heidary. *Spectral norms of independent entries with regular moment growth*. Lean source and verification evidence, pinned commit 762bd5ec5050a96f5e6ba3926b6cda4816fcd4b0. See `Solution.lean`, `MI32/SymmetricDeletion.lean`, and the source-fidelity record; consulted 16 September 2026. [Pinned source repository](#).